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Income predictability and budgeting

C. Yiwei Zhang[1] | **Abigail B. Sussman**[2]

[1]Department of Consumer Science, University of Wisconsin-Madison, Madison, Wisconsin, USA

[2]Department of Marketing, University of Chicago Booth School of Business, Chicago, Illinois, USA

Correspondence
C. Yiwei Zhang, University of Wisconsin-Madison, 1300 Linden Drive, Madison, WI 53706, USA.
Email: cyzhang@wisc.edu

Abstract

Unpredictable income poses a significant threat to economic stability for many U.S. households. While budgeting may help to mitigate some of the negative effects of unpredictable income, the process of budgeting can also be more challenging under such circumstances. This paper presents new descriptive evidence on the relationship between income predictability, financial wellbeing, and budgeting behavior using data from a nationally-representative survey of nearly 4000 U.S. adults. We document a strong negative relationship between income unpredictability and financial wellbeing. However, the strength of this negative relationship is substantially attenuated for those who budget. This suggests that budgeting may be a useful tool for mitigating the adverse effects of unpredictable income. Despite these benefits, the propensity to budget is significantly lower among those with the most unpredictable income. These findings indicate the need for policy reforms that insure households against unpredictable income in addition to efforts targeting individual-level financial management strategies.

KEYWORDS
budgeting, income predictability, income volatility

1 | INTRODUCTION

A significant share of U.S. households experience financial instability, with roughly one-third of households unable to come up with $400 if needed without borrowing or relying on others (Federal Reserve, 2022). A growing literature has documented the important role that unpredictable fluctuations in income can play in contributing to this financial strain (Amorim & Schneider, 2022; Farrell et al., 2019; Pew Charitable Trusts, 2017). Budgeting and similar financial planning approaches may help to mitigate some of the effects of income unpredictability. At the same time, engaging in such practices may also be more challenging for those with highly unpredictable income flows. Despite its potential importance for financial wellbeing, relatively little is known about the role of budgeting for households with unpredictable income.

This paper uses newly available cross-sectional data from a survey we administered to a nationally-representative panel of nearly 4000 U.S. adults to study the relationship between income predictability, financial wellbeing, and budgeting behavior. Using these data, we show that households with unpredictable incomes report being substantially less well-off financially. This relationship persists across four different measures of both objective and subjective financial wellbeing and holds even conditional on income, pay frequency, and a host of demographic characteristics. Importantly, however, the strength of the relationship between lower income predictability and lower financial wellbeing is considerably attenuated for respondents who budget. For example, the gap in perceived financial wellbeing between those with predictable and unpredictable incomes shrinks by over 25% among households that choose to budget. While our estimates are not causal, these findings nonetheless suggest that individuals that experience significant income predictability may benefit from engaging in budgeting.[1]

Despite these potential benefits, when we consider the association between income predictability and the propensity to budget, we find that respondents with lower income predictability are significantly *less* likely to report that they budget. Relative to those with the highest income predictability, respondents with the lowest income predictability are 14 percentage points less likely to report that they currently budget, after controlling for key characteristics. This difference is large and represents a 20% lower likelihood of budgeting relative to a baseline likelihood of 71% for those with the highest income predictability. One possible reason for this disparity is that budgeting is difficult when income is unpredictable. Consistent with this, we find that those with the lowest income predictability are significantly more likely to report that they find budgeting to be unpleasant.

Taken together, these findings highlight a fundamental challenge for practitioners and policymakers as they consider how to help households manage their finances in the face of growing economic uncertainty. Our research indicates that the strong negative relationship between income unpredictability and financial wellbeing is significantly attenuated for those who budget, and yet, we also find a lower propensity to budget among those with the most unpredictable income. In other words, those who might benefit most from budgeting are also the ones least likely to engage in budgeting. These findings indicate the potential need for more large-scale policy reforms focused on insuring households against unpredictable income flows rather than more targeted interventions focused on individual-level financial management strategies.

Our paper contributes to a large literature studying the nature and consequences of household income volatility. Early studies in this literature often focused on examining long-run trends and estimated income volatility using year-to-year or biennial changes in earnings

(Dynan et al., 2012; Gottschalk & Moffitt, 1994; Moffitt et al., 2023). More recently, interest has expanded to include higher frequency fluctuations in income that happen on a month-to-month or even week-to-week basis within the year (Farrell et al., 2019; Morduch & Schneider, 2017).

Our paper differs from and contributes to this literature in two key ways. First, by explicitly asking respondents whether they can predict their income, our survey design allows us to focus specifically on the portion of income volatility that households view to be *unpredictable*. Income volatility can stem from both predictable and unpredictable fluctuations in income. This distinction is important because predictable fluctuations in income are, in theory, more easily planned around than unpredictable fluctuations in income. For example, a teacher who experiences a predictable shortfall in their income over the summer faces different challenges than a retail worker whose income varies unpredictably because their hours of scheduled work are uncertain.[2] While households experiencing predictable income volatility may of course still face constraints that make income swings challenging (e.g., limited access to credit, inability to take on a second job, etc.), those challenges would likely be even greater were the income fluctuations unpredictable.

Second, while several existing papers document relationships between income volatility and various measures of household wellbeing (Bartfeld & Collins, 2017; Farrell & Greig, 2015; Farrell & Greig, 2017; Fisher, 2010; Peetz & Robson, 2019; Pew Charitable Trusts, 2017; Schneider & Harknett, 2019; West & DeVoe, 2021), ours is the first to focus explicitly on how these relationships are moderated by household budgeting practices. The survey we administer was designed specifically to measure budgeting behavior and contains questions often not available in existing survey or administrative data sources used to study income volatility. Our findings here highlight both the promise and limitations of budgeting as a tool for weathering unpredictable fluctuations in income.

2 | DATA AND METHODOLOGY

The central contribution of our paper is the collection and analysis of a new dataset that allows us to examine the relationship between income predictability, financial wellbeing, and the budgeting process. In this section, we provide additional background on this data and describe our specific measure of income predictability and how it compares with similar survey-measures of income volatility. We then detail our empirical approach.

2.1 | Budgeting data

We draw on cross-sectional data from a survey we administered to a nationally-representative panel of U.S. adults through Lucid, an online platform used to carry out surveys and experiments. In addition to basic demographic and income information, the survey data captures detailed information on budgeting behaviors and beliefs for nearly 4000 individuals. For our analysis, we focus on several key questions from the survey. We consider respondents' experience with budgeting, whether formally or informally, and their reasons for budgeting or not budgeting.[3] We also examine how unpleasant respondents consider budgeting to be and why they consider it unpleasant if they do. To better understand the specific budgeting practices that respondents engage in, we consider detailed questions regarding how they set and manage their budgets. Finally, we also assess the financial wellbeing of respondents and whether they believe budgeting will help them make more financially prudent decisions.

In Table A1 (replicated from Zhang et al., 2022), we benchmark the representativeness of our sample against the full U.S. population using data from the American Community Survey (ACS) and the Current Employment Statistics (CES) survey. The results from this comparison show that our sample is broadly representative of the U.S. population though respondents to our survey appear to be slightly higher income and more educated. Zhang et al. (2022) provides more information on the full set of survey questions as well as detailed descriptions of respondent recruitment and validity checks.

Table 1 presents descriptive statistics on key demographic and economic characteristics for our sample. About 54% of respondents in our survey are female. Roughly 57% live with either a spouse or partner, and 39% have at least one child in their household. Survey respondents also span the socio-economic spectrum. While nearly 37% of respondents had at least a bachelor's degree, roughly 25% of respondents had a high-school degree or less. About 56% of respondents are currently working, either part-time or full-time. Roughly 18% of respondents reported an annual household income less than $20,000 while 31% had an annual household income greater than $70,000.

2.2 | Measuring income predictability

To measure income predictability, we asked survey respondents to report how predictable their income is from month to month on a Likert scale ranging from 1 ("Completely Unpredictable") to 7 ("Completely Predictable"). Figure 1 plots the distribution of income predictability scores reported by respondents. As the figure shows, the distribution is highly skewed with an average score of 5.3. Roughly 75% report their income to be fairly predictable from month to month (score of 5, 6, or 7) with about 30% indicating that their income is completely predictable (score of 7). The remaining 25% (score of 4 or lower) report experiencing some degree of income unpredictability, with 5% reporting their income is completely unpredictable from month to month (score of 1).

Unlike our measure, most other existing survey-based measures of self-reported income changes focus on measuring income volatility and do not make an explicit distinction between whether volatility is predictable versus unpredictable. Because of this, it is not possible to make a perfect comparison of our measure of income predictability and related questions on income volatility from such surveys. However, the two measures nonetheless appear to be fairly consistent. To show this, we focus on income volatility measures from two publicly available surveys: (1) the 2018 Survey of Household Economics and Decision-making (SHED) administered by the Federal Reserve Board and (2) the National Financial Well-Being Survey administered by the Consumer Financial Protection Bureau. While other surveys have fielded similar questions on income volatility, the SHED and the National Financial Well-Being Survey have the advantages of being sampled to be nationally representative of U.S. adults, similar to our survey. To measure income volatility, the SHED asks respondents: "In the past 12 months, which one of the following best describes your income?" Respondents are able to indicate one of three possible responses: (1) Roughly the same amount each month, (2) Occasionally varies from month to month, and (3) Varies quite often from month to month. The National Financial Well-Being Survey asks a similar question: "Which of the following best describes how your household's income changes from month to month, if at all?" Respondents are able to indicate one of three

TABLE 1 Summary statistics.

Panel A. Demographic characteristics	Share of respondents (%)
Age bracket	
18–24 years	10.2
25–34 years	20.9
35–44 years	19.8
45–54 years	17.7
55+ years	31.4
Gender	
Female	53.5
Male	46.5
Employment status	
Student	4.2
Part-time	14.2
Full-time	42.1
Not currently employed	39.4
Education	
High-school degree or less	24.8
Some college or Associate's degree	38.3
Bachelor's degree or more	36.9
Marital status	
Lives with spouse/partner	57.1
Does not live with spouse/partner	42.9
Children in the household	
Does not have children	60.7
Has children	39.3

Panel B. Financial characteristics	Share of respondents (%)
Household income	
<20 K	17.7
20 K–30 K	12.8
30 K–40 K	11.4
40K–50 K	9.7
50 K–60 K	10.5
60 K–70 K	7.3
>70 K	30.7
Pay frequency	
Weekly	19.0

TABLE 1 (Continued)

Panel B. Financial characteristics	
	Share of respondents (%)
Biweekly	24.9
Semi-monthly	14.2
Monthly	31.5
Irregular frequency	5.5
Number of participants	3826

Note: This table presents descriptive statistics for the full sample of survey respondents. Table entries provide the frequency breakdown of key demographic and financial characteristics in percentage terms. The count of respondents is listed in the final row.

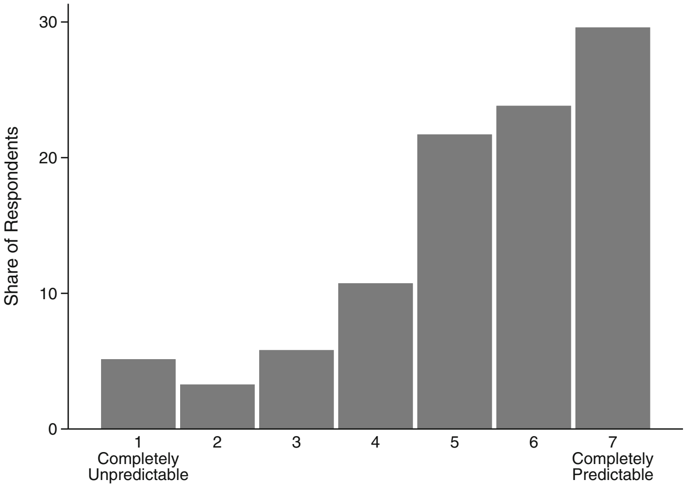

FIGURE 1 Distribution of income predictability scores. This figure plots the distribution of income predictability scores for the full sample of survey respondents. Scores were given in response to a question on how predictable respondents' income is from month to month and range on a scale ranging from 1 ("Completely Unpredictable") to 7 ("Completely Predictable").

possible responses: (1) Roughly the same each month, (2) Roughly the same most months, but some unusually high or low months during the year, and (3) Often varies quite a bit from one month to the next. Similar to our measure of income predictability, the evidence from these two surveys indicates that the distribution of income volatility is highly skewed. In both surveys, approximately 70% of respondents report their income to be roughly the same each month. The remaining 30% report experiencing some degree of income volatility, with roughly 10% reporting that their income varies quite a bit from month to month.

Table 2 shows how our measure of income predictability varies by key characteristics. Column (1) of Table 2 reports the average score by the demographic and economic characteristics presented in Table 1. Columns (2) to (5) of Table 2 report the share of respondents in each quartile of income predictability for the indicated set of respondents. For example, whereas 45% of respondents between 18 and 24 years of age are in the first quartile of income predictability (i.e., lowest levels of income predictability), only 14% of respondents over 55 fall within that same quartile.

TABLE 2 Income predictability by individual characteristics.

	(1)	Share of respondents (%) (2)	(3)	(4)	(5)
	Mean income predictability score	Quartile 1 (Score 1–4)	Quartile 2 (Score 5)	Quartile 3 (Score 6)	Quartile 4 (Score 7)
Overall sample	5.30	0.25	0.22	0.24	0.30
Panel A. Demographic characteristics					
Age bracket					
18–24 years	4.54	44.9	24.1	15.6	15.4
25–34 years	5.01	29.1	27.5	22.6	20.8
35–44 years	5.16	27.4	25.5	23.4	23.7
45–54 years	5.31	24.2	21.9	24.0	29.9
55+ years	5.83	14.5	14.5	27.4	43.6
Gender					
Female	5.23	26.7	21.4	23.1	28.8
Male	5.39	22.8	22.0	24.7	30.5
Employment status					
Student	4.59	45.0	25.6	11.3	18.1
Part-time	4.72	37.8	28.6	18.6	14.9
Full-time	5.35	21.8	25.5	27.0	25.7
Not currently employed	5.55	21.3	14.8	23.6	40.2
Education					
High-school degree or less	5.05	31.4	22.7	18.0	28.0
Some college or Associate's degree	5.22	27.2	20.9	24.0	27.9
Bachelor's degree or more	5.56	18.2	21.8	27.6	32.4
Marital status					
Lives with spouse/partner	5.20	29.2	20.1	20.5	30.2
Does not live with spouse/partner	5.39	21.7	22.9	26.3	29.1
Children in the household					
Does not have children	5.35	25.5	19.5	22.1	33.0
Has children	5.23	24.0	25.2	26.4	24.4
Panel B. Financial characteristics					

TABLE 2 (Continued)

	(1) Mean income predictability score	Share of respondents (%) (2) Quartile 1 (Score 1–4)	(3) Quartile 2 (Score 5)	(4) Quartile 3 (Score 6)	(5) Quartile 4 (Score 7)
Household income					
<20 K	4.90	38.3	15.5	14.9	31.3
20 K–30 K	5.10	28.0	24.3	20.9	26.8
30 K–40 K	5.22	28.5	20.9	25.8	24.8
40K–50 K	5.22	25.6	26.2	23.2	25.1
50 K–60 K	5.39	22.0	23.3	24.5	30.3
60 K–70 K	5.37	21.4	26.3	24.9	27.4
>70 K	5.63	16.2	21.4	29.2	33.3
Pay frequency					
Weekly	5.04	30.7	26.7	22.0	20.6
Biweekly	5.29	23.5	26.2	26.1	24.1
Semi-monthly	5.44	19.6	25.5	29.0	26.0
Monthly	5.86	14.5	14.4	25.2	45.9
Irregular frequency	3.62	66.0	16.3	7.2	10.5

Note: This table presents summary statistics on how income predictability varies with key demographic and economics characteristics. Column (1) reports the average income predictability score by these characteristics for the full sample of survey respondents. Columns (2) to (5) of Table 2 report the share of respondents in each quartile of income predictability for the indicated set of respondents in the row.

In general, we find that income predictability varies with these characteristics as we might expect. Income predictability scores are monotonically increasing in age, education, and income. When looking at employment status, students report the lowest levels of income predictability with a mean score of 4.59. Those working part-time report lower income predictability than those working full-time. Respondents who are not currently employed report the highest levels of income predictability with an average score of 5.55. This relatively high degree of income predictability may in part be due to the inclusion of non-working adults in single-income households or retirees and other individuals who receive regular benefit payments. Those paid more infrequently (e.g., monthly) experience greater income predictability than those paid more frequently (e.g., weekly). Unsurprisingly, those who report being paid at irregular frequencies also report the lowest levels of income predictability. We also find that being male, living with a spouse or partner, and not having children are associated with higher levels of income predictability.[4]

2.3 | Empirical approach

We are interested in examining the relationship between income predictability and budgeting. To motivate, we first consider how financial wellbeing correlates with income predictability and whether this relationship differs for those who do versus do not budget. We then examine whether the propensity to budget and the reasons for budgeting or not vary with income predictability.

For each outcome we consider, we first show how its unconditional distribution varies by quartile of income predictability, both for the overall sample of respondents and separately by household income. This allows us to examine the unconditional relationship between the outcome of interest and income predictability as well as examine how that relationship may differ for those with relatively low versus high household income. To allow for the cleanest interpretation, we focus on comparisons between those with the least versus most predictable incomes (i.e., those falling within the lowest versus highest quartile of income predictability).

After presenting evidence on the unconditional relationship, we then report regression-adjusted results that take into account the correlation between income predictability and key characteristics. In particular, we estimate versions of the following specification:

$$Y_i = \alpha + X_i'\gamma + \beta_1 \times Quartile1_i + \beta_2 \times Quartile2_i + \beta_3 \times Quartile3_i + \epsilon_i \tag{1}$$

where Y_i is the outcome of interest for individual i, X_i is a vector of individual-level demographics (age, gender, education, employment status), household structure (marital status, number of children in the household), and financial observables (household income and pay frequency). These latter two controls for income and pay frequency are particularly important, as including them means that any relationship we document between income predictability and household outcomes holds even conditional on the overall level of income and any predictable volatility in income arising from variation in pay frequencies. The indicator variables $Quartile1_i$, $Quartile2_i$, and $Quartile3_i$ take the value one if individual i reports their income being in the first, second, or third quartile of predictability, respectively. The coefficients of interest are β_1, β_2, and β_3, which measure the difference in average outcomes for individuals in the indicated quartile of predictability relative to that of respondents with the highest income predictability (i.e., in the fourth quartile), holding constant individual characteristics.

3 | RESULTS

3.1 | Lower income predictability is associated with lower financial wellbeing

We first examine the relationship between income predictability and overall financial wellbeing. We proxy for financial wellbeing using four different measures: asset holdings, the amount saved last month, confidence in the ability to come up with funds if an unexpected need arose, and perceived wellbeing. The first two proxies are objective measures of a respondent's financial state whereas the latter two proxies capture more subjective experiences of financial wellbeing.

Across all four measures, we find evidence that lower income predictability is associated with significantly lower financial wellbeing. We focus first on the relationship between asset holdings and income predictability. Figure 2 presents the distribution of asset levels by quartile of income predictability across several income ranges. The variable for asset levels takes on 8 discrete values: \$0–\$249, \$250–499, \$500–\$999, \$1000–\$4999, \$5000–\$9999, \$10,000–\$99,999, and \$100,000+. For ease of reference, we group these values into bins. Two empirical patterns

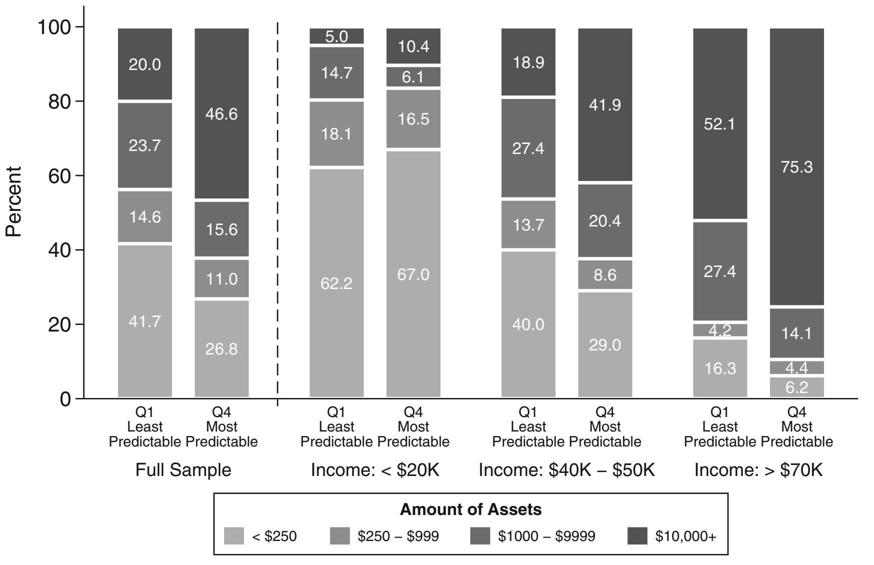

FIGURE 2 Asset levels by income predictability. This figure plots the distribution of asset levels for those in the lowest (Q1) versus highest (Q4) quartile of income predictability for the full sample of survey respondents, as well as across three subsamples based on household income ranges: less than $20,000, between $40,000 to $50,000, and greater than $70,000.

are worth noting. First, lower income predictability is associated with lower levels of asset accumulation. Over 40% of respondents in the lowest quartile of income predictability have less than $250 of assets in comparison with only 27% of those in the highest quartile of income predictability. Similarly, roughly 20% of those in the lowest quartile of income predictability have over $10,000 in assets in comparison with 47% of those in the highest quartile. Second, while one might expect that income unpredictability is most problematic for those with the lowest incomes, we find instead that this difference in asset levels between those in the lowest versus highest quartile of income predictability is driven by those with *higher* levels of income. For respondents with less than $20,000 in income, the difference in asset levels appears to be relatively small. However, this gap grows significantly as income increases.

These patterns are robust to controlling for key demographic and economic characteristics. Table 3 presents both unadjusted estimates corresponding to Figure 2 (column (1)) and regression-adjusted estimates controlling for demographics, household structure, income, and pay frequency (column (2)), with reported asset levels as the outcome of interest. Because respondents were asked to indicate the asset level range they belong to, rather than provide a precise value, we use an interval regression that accounts for both the interval nature of the outcome as well as the right-censoring of asset levels in the highest range.[5] The estimate for "Constant" in the bottom row gives the average baseline amount of assets for respondents in the highest quartile of income predictability. The remaining estimates represent the relative difference in assets for respondents in the indicated quartile of predictability.

We find that lower levels of income predictability are associated with significantly lower levels of asset accumulation even after controlling for key characteristics. Relative to respondents with the highest income predictability, those with the lowest income predictability (Quartile 1) hold roughly $10,400 less in assets on average. This represents nearly 25% of the

TABLE 3 Asset levels by income predictability.

	(1)	(2)
Quartile 1	−27,101.90***	−10,424.21***
	(2028.79)	(1778.99)
Quartile 2	−15,971.92***	−8097.98***
	(2115.80)	(1781.69)
Quartile 3	−2688.18	−2698.83
	(2069.86)	(1696.15)
Constant	42,173.48***	36,257.04***
	(1392.46)	(1166.53)
Demographic controls		Y
Household structure controls		Y
Income controls		Y
N	3777	3777

Note: This table presents estimates of the differential in assets for respondents in the first, second, or third quartile of income predictability relative to those in the highest quartile of income predictability (i.e., those with the most predictable incomes). Each column reports estimates from an interval regression where the dependent variable is the reported asset level. Coefficients are reported for a "Quartile 1" indicator, "Quartile 2" indicator, and "Quartile 3" indicator, denoting whether a respondent is among those in the first, second, or third quartile of income predictability, respectively. The estimate for "Constant" in the bottom row gives the average baseline amount of assets for the omitted group of respondents in the highest quartile of income predictability (Quartile 4). Column (1) presents unadjusted estimates, and Column (2) presents regression-adjusted estimates that control for age, gender, education, employment status, marital status, number of children in the household, household income, and pay frequency. Significance levels 10%, 5%, 1%, and 0.1% are denoted by +, *, **, and ***, respectively.

unconditional average level of assets among those with the highest level of income predictability.

We find a similar relationship between predictability and financial wellbeing when examining how the amount that respondents report saving in the prior month varies with income predictability. Figure 3 plots the distribution of amount saved in the prior month by income predictability quartile for the full sample as well as across income ranges. The variable for savings levels takes on 6 discrete values: $0, $1–$49, $50–$99, $100–$499, $500–$999, and $1000+. For ease of reference, we group these values into bins. Consistent with our findings on asset levels, we see that lower income predictability is associated with lower levels of savings. Roughly 64% of respondents in the highest quartile of income predictability were able to save at least $100 in the prior month in comparison with only 41% of those in the lowest quartile of predictability. And although 8% of respondents in the highest quartile of income predictability did not save at all in the prior month, nearly twice as many respondents in the lowest quartile of predictability were unable to save. We once again find that differences in the ability to save by income predictability are generally highest for those with the highest income levels in our sample and this relative gap decreases as income decreases.

Table 4 presents both unadjusted estimates (column (1)) and regression-adjusted estimates controlling for demographics, household structure, income, and pay frequency (column (2)), with the amount saved in the prior month as the outcome of interest. We once again use an interval regression to account for both the interval nature of the outcome as well as the right-censoring of savings amounts in the highest range.[6] The estimate for "Constant" in the bottom

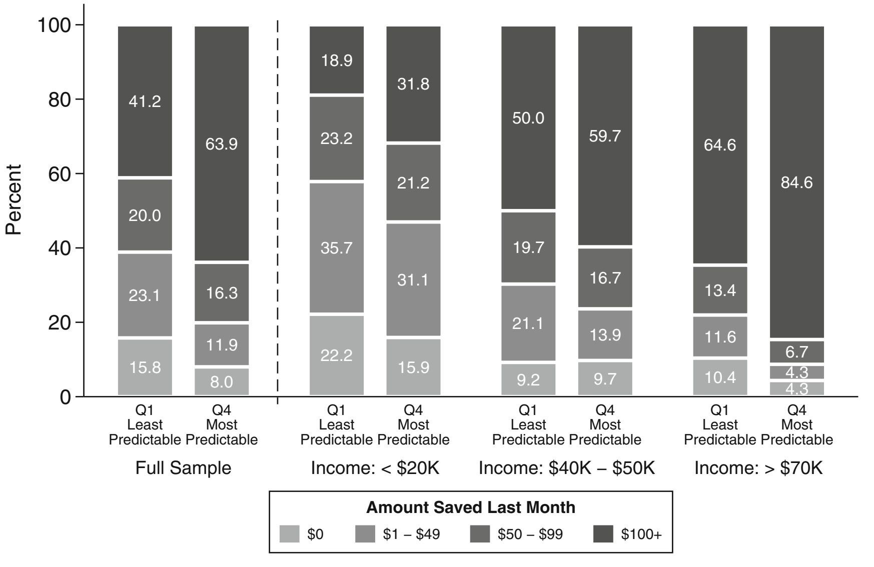

FIGURE 3 How Much Saved Last Month by Income Predictability. This figure plots the distribution of amount saved in the prior month for those in the lowest (Q1) versus highest (Q4) quartile of income predictability for the full sample of survey respondents, as well as across three subsamples based on household income ranges: less than $20,000, between $40,000 to $50,000, and greater than $70,000.

row gives the average baseline amount saved in the prior month by respondents in the highest quartile of income predictability. The remaining estimates represent the relative difference in savings for respondents in the indicated quartile of predictability.

Consistent with Figure 3, we find that lower levels of income predictability are associated with significantly lower levels of savings, even after controlling for key characteristics. Relative to respondents with the highest income predictability, those with the lowest income predictability (Quartile 1) were able to save roughly $93 less on average in the prior month.

We next consider the relationship between confidence in the ability to come up with funds in the event of an unexpected need and income predictability. Confidence is measured on a four-point scale that ranges from "1: I am certain I cannot come up with ($X if an unexpected need arose)" to "4: I am certain I could come up with ($X if an unexpected need arose)" with respondents presented with a range of dollar amounts: $10, $25, $50, $100, $200, $500, $1000, $2000, and $5000. Figure 4 plots the distribution of responses for respondents' confidence in the ability to come up with $500 by whether a respondent is in either the lowest or highest income predictability quartile for the full sample and across three different income ranges.

As the figure shows, regardless of income, lower income predictability is associated with lower reported confidence in the ability to come up with $500 if an unexpected need arose. Also consistent with our prior findings, the relative gap in confidence between those in the lowest versus highest quartile of income predictability is increasing in household income. For respondents with less than $20,000 in income, those with the least predictable income are 5.5 percentage points less likely than those in the highest quartile of income predictability to report that they probably or certainly can come up with $500 if an unexpected need arose. For respondents with greater than $70,000 income, this difference nearly triples to 15.5 percentage points.

TABLE 4 How much saved last month by income predictability.

	(1)	(2)
Quartile 1	−171.18***	−92.51***
	(17.14)	(16.52)
Quartile 2	−95.91***	−75.07***
	(17.64)	(16.25)
Quartile 3	−40.96*	−45.33**
	(17.12)	(15.27)
Constant	378.28***	356.63***
	(11.70)	(10.73)
Demographic controls		Y
Household structure controls		Y
Income controls		Y
N	3049	3049

Note: This table presents estimates of the differential in the amount saved in the prior month for respondents in the first, second, or third quartile of income predictability relative to those in the highest quartile of income predictability (i.e., those with the most predictable incomes). Each column reports estimates from an interval regression where the dependent variable is the reported savings level. Coefficients are reported for a "Quartile 1" indicator, "Quartile 2" indicator, and "Quartile 3" indicator, denoting whether a respondent is among those in the first, second, or third quartile of income predictability, respectively. The estimate for "Constant" in the bottom row gives the average baseline amount saved in the prior month for the omitted group of respondents in the highest quartile of income predictability (Quartile 4). Column (1) presents unadjusted estimates, and Column (2) presents regression-adjusted estimates that control for age, gender, education, employment status, marital status, number of children in the household, household income, and pay frequency. Significance levels 10%, 5%, 1%, and 0.1% are denoted by +, *, **, and ***, respectively.

Table 5 presents a series of regression-adjusted estimates controlling for demographics, household structure, and income. For each column, the outcome of interest is the reported confidence in the ability to come up with the indicated amount of funds if an unexpected need arose. As noted above, confidence is measured on a four-point scale with higher scores corresponding to higher levels of confidence.[7] The estimate for "Constant" in the bottom row gives the average baseline confidence for respondents in the highest quartile of income predictability. The remaining estimates represent the relative difference in confidence for respondents in the indicated quartile of predictability.

First, as we might expect, Table 5 shows that overall confidence decreases as the amount of funds required if an unexpected need arose increases. Comparing columns (1) and (4) for the highest quartile of income predictability, we see that the overall confidence in the ability to come up with $100 is nearly 1.6 times as high as for $5000 and that this confidence declines monotonically across columns. Second, we find that the patterns we observe in Figure 4 persist even after controlling for key demographic and economic characteristics. Those in the lowest quartile of income predictability (Quartile 1) consistently report significantly lower levels of confidence in their ability to come up with funds if an unexpected need arose relative to those in the highest quartile of predictability.

Finally, we assess how a respondent's perceived wellbeing varies with the predictability of their income. To do so, we focus on responses to four questions: (1) I will achieve the financial goals that I have set for myself; (2) I have saved (or will be able to save) enough money to last

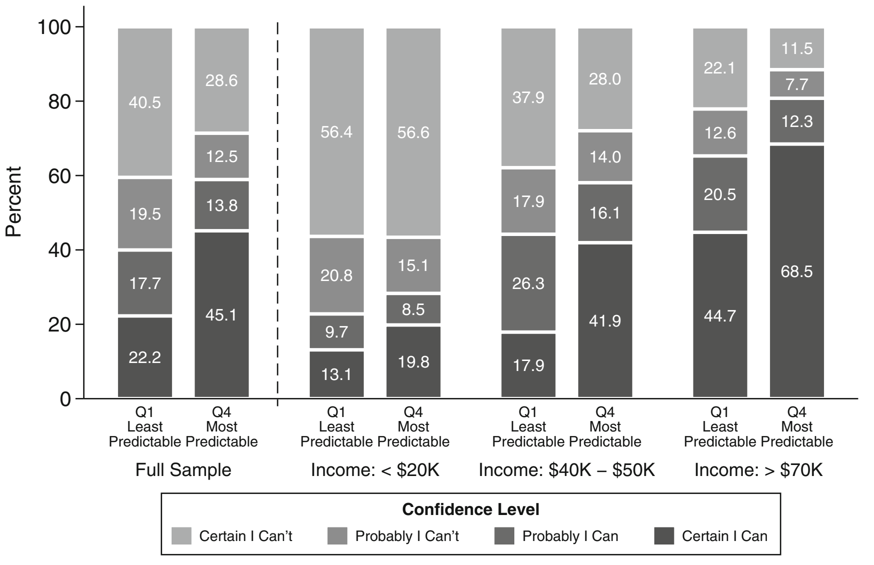

FIGURE 4 Confidence in ability to come up with $500 by income predictability. This figure plots the distribution of responses for respondents' confidence in the ability to come up with $500 for those in the lowest (Q1) versus highest (Q4) quartile of income predictability for the full sample of survey respondents, as well as across three subsamples based on household income ranges: less than $20,000, between $40,000 to $50,000, and greater than $70,000.

me to the end of my life; (3) I am behind on my finances; and (4) Because of my money situation, I feel I will never have the things I want in life. Responses for each question are measured using a five-point scale that ranges from "1: Describes me not at all" to "5: Describes me completely." The first two questions are designed to capture respondents' expectations about their future financial security whereas the latter two questions are designed to capture respondents' stress about their current money management (CFPB, 2017; Netemeyer et al., 2018). For our main analysis, we create a composite index of perceived wellbeing by combining the individual scores from these four questions.[8] The index ranges in value from −8 to 8, with higher values indicating higher perceived wellbeing.

Figure 5 plots the distribution of this perceived wellbeing index by whether a respondent is in either the lowest or highest income predictability quartile for the full sample, as well as across three different income ranges. We find that lower income predictability is consistently associated with lower perceived wellbeing, regardless of income level. When looking at the full sample, nearly half of respondents in the highest quartile of income predictability report relatively high perceived wellbeing scores (between 3 and 8) in comparison with only about one-fifth of respondents in the lowest quartile of predictability. And while respondents with lower income generally report lower perceived wellbeing than those with higher income, we further find that the relative difference in perceived wellbeing between those in the lowest versus highest quartile of income is increasing in household income.

These patterns are robust to controlling for key demographic and economic characteristics. Table 6 presents both unadjusted estimates (column (1)) and regression-adjusted estimates controlling for demographics, household structure, income, and pay frequency (column (2)), with our perceived wellbeing index as the outcome of interest. The estimate for "Constant" in the bottom row gives the average perceived wellbeing index score for respondents in the highest

TABLE 5 Confidence in ability to come up with funds by income predictability.

	(1)	(2)	(3)	(4)
	$100	**$500**	**$1000**	**$5000**
Quartile 1	−0.27***	−0.34***	−0.33***	−0.32***
	(0.05)	(0.05)	(0.05)	(0.05)
Quartile 2	−0.06	−0.18***	−0.19***	−0.24***
	(0.05)	(0.05)	(0.05)	(0.05)
Quartile 3	0.05	0.02	−0.04	−0.11*
	(0.04)	(0.05)	(0.05)	(0.05)
Constant	3.29***	2.73***	2.46***	2.06***
	(0.03)	(0.03)	(0.03)	(0.03)
Demographic controls	Y	Y	Y	Y
Household structure controls	Y	Y	Y	Y
Income controls	Y	Y	Y	Y
N	3781	3781	3781	3781

Note: This table presents estimates of the differential confidence in the ability to come up with funds for respondents in the first, second, or third quartile of income predictability relative to those in the highest quartile of income predictability (i.e., those with the most predictable incomes). Each column reports estimates from an OLS regression where the dependent variable is the reported confidence in the ability to come up with the amount of funds indicated in the column heading if an unexpected need arose. Controls include age, gender, education, employment status, marital status, number of children in the household, household income, and pay frequency. Coefficients are reported for a "Quartile 1" indicator, "Quartile 2" indicator, and "Quartile 3" indicator, denoting whether a respondent is among those in the first, second, or third quartile of income predictability, respectively. The estimate for "Constant" in the bottom row gives the average baseline confidence for the omitted group of respondents in the highest quartile of income predictability (Quartile 4). Significance levels 10%, 5%, 1%, and 0.1% are denoted by +, *, **, and ***, respectively.

quartile of income predictability. The remaining estimates represent the relative difference in score for respondents in the indicated quartile of predictability. As Table 6 shows, we find that lower levels of income predictability are associated with significantly lower perceived wellbeing, even after controlling for key characteristics.

Across all proxies, we find consistent evidence that low income predictability is associated with significantly lower levels of financial wellbeing. Beyond this overall finding, one additional empirical pattern worth highlighting is that the relative gap in financial wellbeing between those with low versus high income predictability is often driven by higher income households. In some instances, there are only marginal differences in wellbeing by income predictability for the lowest-income households. One possible explanation for this is that for lower-income households, income predictability may play a smaller role in determining financial wellbeing than a household's general lack of resources. This suggests that practitioners and policymakers may need to carefully consider how they prioritize their efforts to support financially vulnerable populations. We view the interaction between income predictability and the overall availability of financial resources to be an important area of future research.

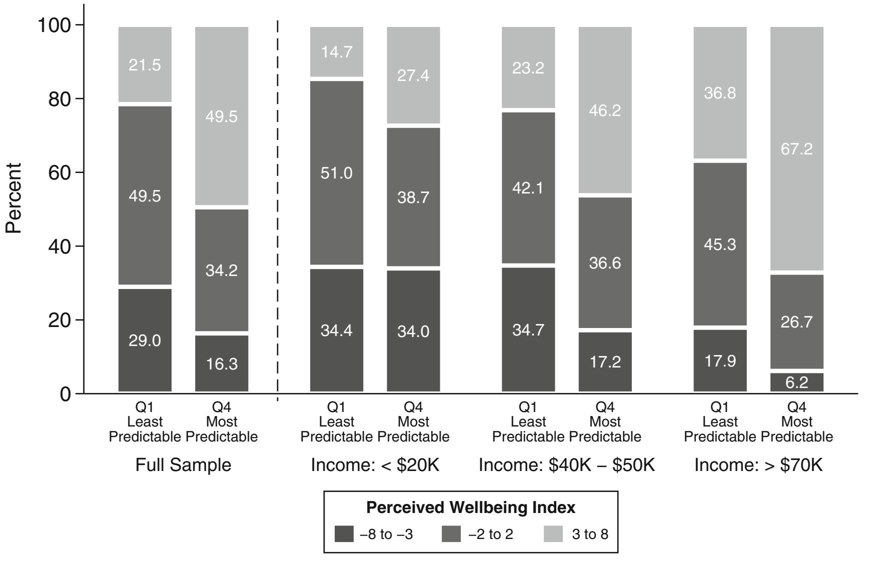

FIGURE 5 Perceived financial wellbeing by income predictability. This figure plots the distribution of perceived wellbeing index scores for those in the lowest (Q1) versus highest (Q4) quartile of income predictability for the full sample of survey respondents, as well as across three subsamples based on household income ranges: less than \$20,000, between \$40,000 to \$50,000, and greater than \$70,000. The index ranges in value from −8 to 8, with higher values indicating higher perceived wellbeing, and is a composite score created by combining individual scores to four questions designed to capture respondents' expectations about their future financial security and stress about their current money management. Additional details on the construction of the perceived wellbeing index can be found in Appendix D (Figures D1, D2; Table D1).

3.2 | Budgeting moderates the relationship between income predictability and financial wellbeing

In the previous section, we documented the strong relationship between income predictability and financial wellbeing. Households in the highest quartile of income predictability have substantially higher levels of financial wellbeing than those in the lowest quartile. Budgeting and similar financial planning approaches may help to mitigate some of the negative effects associated with having unpredictable income.

In this section, we examine the extent to which budgeting may moderate the relationship between income predictability and financial wellbeing. To do so, we estimate the following modified version of the baseline specification from equation (1) for each of our four proxy measures of financial wellbeing:

$$Y_i = \alpha + X_i\gamma + \beta_1 \times \text{Budgets}_i + \beta_2 \times \text{Predictable}_i + \beta_3 \times \text{Predictable}_i \times \text{Budgets}_i + \epsilon_i \qquad (2)$$

This regression is the same as the baseline specification in equation (1), with two exceptions. First, rather than including each quartile of the income predictability distribution as a separate regressor, we include a single variable, *Predictable$_i$*, which takes on the values one through four and denotes the quartile of income predictability that individual i reports their income belonging within. The coefficient on this term measures the average change in financial wellbeing associated with a one quartile increase in income predictability. Second, we fully interact this

TABLE 6 Perceived financial wellbeing by income predictability.

	(1)	(2)
Quartile 1	−2.49***	−2.05***
	(0.17)	(0.17)
Quartile 2	−1.08***	−1.03***
	(0.17)	(0.17)
Quartile 3	−0.34*	−0.45**
	(0.17)	(0.16)
Constant	2.11***	2.02***
	(0.11)	(0.11)
Demographic controls		Y
Household structure controls		Y
Income controls		Y
N	3781	3781

Note: This table presents estimates of the differential in perceived wellbeing for respondents in the first, second, or third quartile of income predictability relative to those in the highest quartile of income predictability (i.e., those with the most predictable incomes). Each column reports estimates from an OLS regression where the dependent variable is the respondent's perceived wellbeing index score. Coefficients are reported for a "Quartile 1" indicator, "Quartile 2" indicator, and "Quartile 3" indicator, denoting whether a respondent is among those in the first, second, or third quartile of income predictability, respectively. The estimate for "Constant" in the bottom row gives the average baseline perceived wellbeing index score for the omitted group of respondents in the highest quartile of income predictability (Quartile 4). Column (1) presents unadjusted estimates, and Column (2) presents regression-adjusted estimates that control for age, gender, education, employment status, marital status, number of children in the household, household income, and pay frequency. Significance levels 10%, 5%, 1%, and 0.1% are denoted by +, *, **, and ***, respectively.

income predictability measure with an indicator, $Budgets_i$, which takes the value one if individual i reports that they are currently budgeting. The coefficient of interest on the interaction term, β_3, measures how budgeting moderates the baseline the relationship between income predictability and financial wellbeing.

Table 7 presents estimates from two versions of equation (2) for each measure of financial well-being. In Panel A, we present unadjusted estimates that omit the control variables, X_i'. In Panel B, we regression-adjust these estimates by including controls for demographics, household structure, income, and pay frequency. The coefficient estimate in the first row of each panel represents the baseline relationship between income predictability and financial wellbeing for respondents who do not budget. Consistent with our findings from Section 3.1, we find that lower income predictability is associated with significantly lower financial wellbeing. For example, the estimate in column (2) of Panel B implies that moving from the lowest (first) to highest (fourth) quartile of income predictability is associated with an average increase in savings of $\$38.65 \times (4-1) = \$116$ in the prior month, after controlling for key characteristics.

We further find that the strength of the relationship we observe between income predictability and financial wellbeing is generally attenuated for respondents who budget. For example, the estimate in column (4) of Panel B implies that moving from the lowest (first) to highest (fourth) quartile of income predictability is associated with a $0.73 \times (4-1) = 2.2$ point baseline increase in perceived wellbeing for respondents who are not currently budgeting. This increase

TABLE 7 Income predictability and financial wellbeing by budgeting.

	(1) Amount of assets	(2) Amount saved last month	(3) Confidence in ability to come up with $500	(4) Perceived wellbeing index
Panel A. Without controls				
Predictable	11,524.81***	66.91***	0.26***	0.91***
	(1083.65)	(8.97)	(0.03)	(0.09)
Predictable × budgets	−3276.27*	−21.53+	−0.14***	−0.24*
	(1355.14)	(11.30)	(0.04)	(0.11)
Panel B. With controls				
Predictable	4700.66***	38.65***	0.19***	0.73***
	(915.30)	(8.35)	(0.03)	(0.09)
Predictable × Budgets	−1576.14	−15.44	−0.11***	−0.19+
	(1094.99)	(10.04)	(0.03)	(0.10)
N	3777	3049	3781	3777

Note: This table presents estimates of the relationship between income predictability and financial wellbeing and how it varies by whether or not a respondent budgets. Each column presents a separate regression with the financial wellbeing measure indicated in the column heading as the dependent variable. The first two columns report estimates using an interval regression whereas the remaining estimates report estimates from OLS regressions. Coefficients are reported for a "predictable" variable, which takes on the values one through four and denotes the quartile of income predictability a given individual reports their income belonging within, and its interaction with a "budgets" indicator, which denotes whether the respondent currently budgets. Panel A reports unadjusted estimates whereas Panel B present regression-adjusted estimates that control for age, gender, education, employment status, marital status, number of children in the household, household income, and pay frequency. Significance levels 10%, 5%, 1%, and 0.1% are denoted by +, *, **, and ***, respectively.

is equal to roughly 0.56 standard deviations in the overall index. For respondents who budget, the associated change in perceived wellbeing from moving from the lowest to highest quartile of predictability is reduced by 26%. In general, the magnitude of attenuation is quite large and ranges from a 26% to 58% reduction in the baseline effect depending on the specification or measure of financial wellbeing. While the statistical significance of this moderating effect weakens for some measures of financial wellbeing once we control for key characteristics, the overall effect remains consistent across all proxies. The consistency of these magnitudes suggests that engaging in budgeting may decrease the negative effect of income unpredictability on financial wellbeing. However, the lack of statistical significance leaves this correlation as an open question for future research.

3.3 | Those experiencing lower income predictability are less likely to budget

In the previous section, we documented evidence that suggests budgeting may help attenuate negative effects of income unpredictability on financial wellbeing. However, when we examine how the propensity to budget varies by income predictability, we find that respondents with the lowest income predictability—and who therefore potentially stand to benefit the most from budgeting—are actually the least likely to report that they currently budget.

Figure 6 plots the unconditional likelihood of budgeting, whether formally or informally, by quartile of income predictability for the full sample as well as across several income ranges. We find that respondents' likelihood of budgeting increases in income predictability, with roughly 70% of those in the highest quartile of income predictability reporting that they currently budget in comparison with only 57% of those in the lowest quartile. We also find that this difference in budgeting by income predictability is driven predominantly by differences in the propensity to engage in formal budgeting (e.g., written down or on a website) rather than informal budgeting (e.g., kept in your head).

These patterns are robust to controlling for key demographic and economic characteristics. Table 8 presents both unadjusted estimates (columns (1), (3), and (5)) and regression-adjusted estimates controlling for demographics, household structure, income, and pay frequency (columns (2), (4), and (6)), with an indicator for whether a respondent reports currently budgeting (overall, formally, or informally) as the outcome of interest. The estimate for "Constant" in the bottom row gives the average likelihood a respondent in the highest quartile of income predictability reports currently budgeting. The remaining estimates represent the relative difference in the propensity to budget for respondents in the indicated quartile of predictability.

As Table 8 shows, lower income predictability is associated with a significantly lower likelihood of budgeting. Relative to those with the highest income predictability (Quartile 4), respondents with the lowest income predictability (Quartile 1) are 14 percentage points less likely to report that they currently budget, after controlling for key characteristics. This difference is large and represents a 20% lower likelihood of budgeting relative to a baseline likelihood of 71% for those with the highest income predictability. Consistent with Figure 6, we find that this difference in budgeting by income predictability is driven by differences in the propensity to engage in formal budgeting.

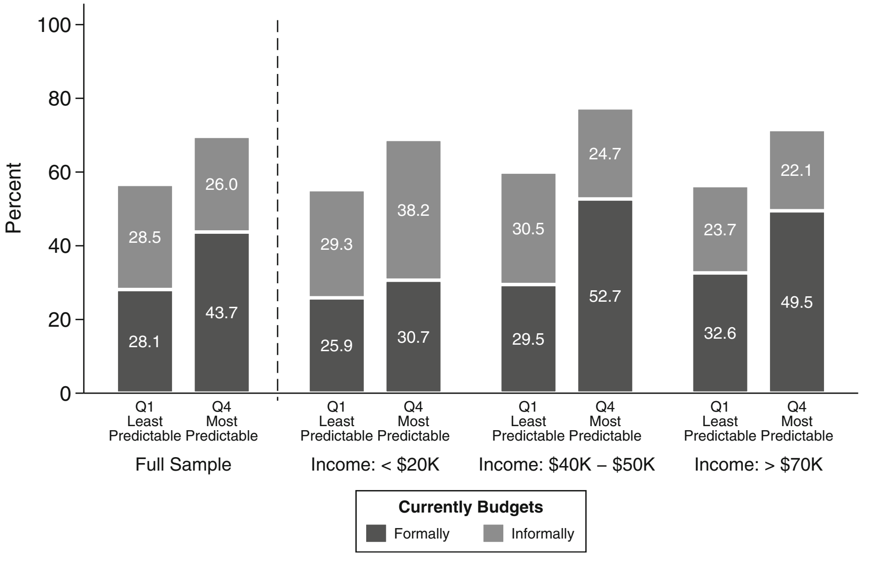

FIGURE 6 Likelihood of budgeting by income predictability. This figure plots the share of respondents who report budgeting, either formally or informally, among those in the lowest (Q1) versus highest (Q4) quartile of income predictability for the full sample of survey respondents, as well as across three subsamples based on household income ranges: less than $20,000, between $40,000 to $50,000, and greater than $70,000.

TABLE 8 Likelihood of budgeting by income predictability.

	Overall		Formally budgets		Informally budgets	
	(1)	(2)	(3)	(4)	(5)	(6)
Quartile 1	−0.13***	−0.14***	−0.16***	−0.17***	0.02	0.03
	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)
Quartile 2	−0.05*	−0.08***	−0.05*	−0.11***	0.00	0.02
	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)
Quartile 3	0.01	−0.01	0.00	−0.04+	0.01	0.03
	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)
Constant	0.70***	0.71***	0.44***	0.46***	0.26***	0.25***
	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)
Demographic controls		Y		Y		Y
Household structure controls		Y		Y		Y
Income controls		Y		Y		Y
N	3781	3781	3781	3781	3781	3781

Note: This table presents estimates of the differential likelihood of budgeting for respondents in the first, second, or third quartile of income predictability relative to those in the highest quartile of income predictability (i.e., those with the most predictable incomes). The first two columns (columns 1 and 2) report estimates from an OLS regression where the dependent variable is an indicator for whether the respondent currently budgets. The remaining four columns report estimates from an OLS regression where the dependent variable is either an indicator for whether the respondent budgets formally (columns 3 and 4) or an indicator for whether the respondent budgets informally (columns 5 and 6). Coefficients are reported for a "Quartile 1" indicator, "Quartile 2" indicator, and "Quartile 3" indicator, denoting whether a respondent is among those in the first, second, or third quartile of income predictability, respectively. The estimate for "Constant" in the bottom row gives the average baseline likelihood of budgeting for the omitted group of respondents in the highest quartile of income predictability (Quartile 4). Columns (1), (3), and (5) present unadjusted estimates, and columns (2), (4), and (6) present regression-adjusted estimates that control for age, gender, education, employment status, marital status, number of children in the household, household income, and pay frequency. Significance levels 10%, 5%, 1%, and 0.1% are denoted by +, *, **, and ***, respectively.

Together, these findings reveal that those with lower income predictability are significantly less likely to engage in budgeting.[9] Given the potential benefits we find to budgeting, it is notable that the very group who could benefit the most are often less likely to budget. In the next section, we explore some of the reasons why those with lower income predictability might choose to not budget.

3.4 | Barriers to budgeting for those with unpredictable income

While budgeting helps to mitigate some of the negative effects of income unpredictability, our results indicate that those with unpredictable income are actually less likely to budget. This may be because having unpredictable income makes budgeting more difficult or unenjoyable or simply because those with unpredictable income are not aware of the benefits that budgeting may entail. In this section, we explore some of the potential barriers to budgeting for those with unpredictable income.

To begin, we first consider respondents' direct motives for not budgeting. As part of our survey, we asked those who are not currently budgeting to identify reasons for why they do not budget. They could select one of several reasons from the following list: (1) I think it's too difficult, (2) I find budgeting unpleasant, (3) I don't think budgeting is useful, (4), I don't think I need to budget because I have enough money. In addition to these response options, respondents could provide their own reasoning through an open-ended response ("other"). Figure 7 presents the distribution of responses to this question for those in the lowest and highest quartiles of income predictability.

We find that the most commonly indicated reasons for not budgeting by those with unpredictable incomes are that it is unpleasant (32.9%) and too difficult (29.3%). Few respondents indicate that they don't believe budgeting to be useful (7.0%) or that they have no need for budgeting (14.3%). In contrast, those with the most predictable incomes are much more likely to indicate that they do not budget because it is not useful (13.1%) or have no need (28.6%) rather than because it is unpleasant (28.6%) or difficult (19.2%). These patterns suggest that for those with highly unpredictable income, the decision not to budget stems not from a lack of recognition of its potential benefits but from more practical challenges that may make budgeting unpleasant or difficult. Consistent with this view, when we ask respondents who do not currently budget whether they think they should keep a budget, nearly 76% of those in the lowest quartile of income predictability believe that they should. By comparison, only 61% of those in the highest quartile of income predictability who are not currently budgeting believe that they should budget.

We further find that the degree of unpleasantness associated with budgeting is particularly acute for those with unpredictable incomes. As part of our survey, we asked respondents to report how unpleasant they find budgeting to be on a seven-point scale, with higher scores corresponding to higher levels of unpleasantness.[10] Figure 8 plots the distribution of responses for the level of unpleasantness that respondents associate with keeping a budget by whether they are in either the lowest or highest income predictability quartile, for the full sample and across three different income ranges. As the figure reveals, lower income predictability is consistently associated with higher levels of unpleasantness in keeping a budget, even for those with relatively high income. Roughly 27% of respondents in the highest quartile of income

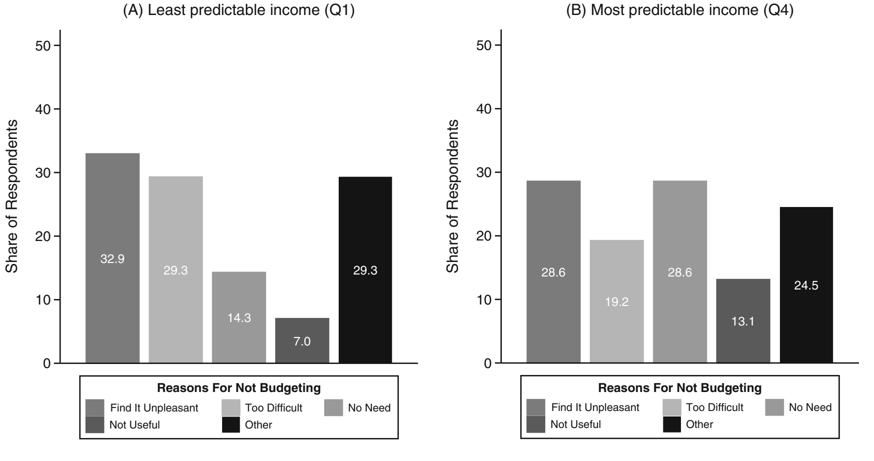

FIGURE 7 Reasons for not budgeting by income predictability. This figure plots the distribution of the share of respondents in the lowest quartile (Panel A) and highest quartile (Panel B) of income predictability who indicate a particular motivation as a main reason for not budgeting for each of the following set of potential motivations: think it's too difficult, find it unpleasant, don't think it's useful, and don't think I need to because I have enough money. In addition to these response options, respondents could provide their own reasoning through an open-ended response ("other").

predictability find keeping a budget to be unpleasant (score of 5, 6 or 7) in comparison with nearly 40% of respondents in the lowest quartile of predictability.

This heightened degree of unpleasantness associated with budgeting for those with unpredictable incomes persists even after accounting for key characteristics. Table 9 presents both unadjusted (column (1)) and regression-adjusted estimates (column (2)) that control for demographics, household structure, income, and pay frequency, with the reported level of unpleasantness as the outcome of interest. As noted above, unpleasantness is measured on a seven-point scale with higher scores corresponding to higher levels of unpleasantness. The estimate for "Constant" in the bottom row gives the average baseline level of unpleasantness for respondents in the highest quartile of income predictability. The remaining estimates represent the relative difference in reported unpleasantness for respondents in the indicated quartile of predictability. We find that those with the lowest income predictability report significantly higher levels of unpleasantness associated with budgeting relative to those with the highest income predictability, even after controlling for key characteristics.

Given that those with lower income predictability report higher levels of unpleasantness associated with budgeting, we next consider why respondents find budgeting to be unpleasant. As part of our survey, we asked respondents who report high levels of unpleasantness (score of 5, 6 or 7) to identify their main reasons for finding budgeting unpleasant. Respondents could select one of multiple reasons from the following list: (1) it takes too much time, (2) it's too hard to follow a budget long-term, (3) budgeting makes me feel like I have less money than I thought I did, and (4) budgeting doesn't help me reach my financial goals. In addition to these response options, respondents could also provide their own reasoning through an open-ended response ("other").

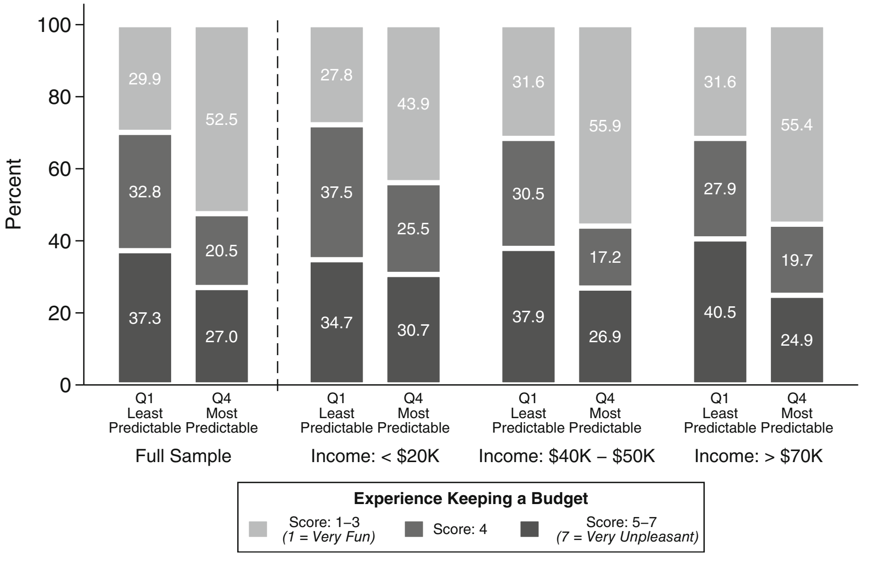

FIGURE 8 Unpleasantness of budgeting by income predictability. This figure plots the distribution of responses for the level of unpleasantness associated with budgeting for those in the lowest (Q1) versus highest (Q4) quartile of income predictability for the full sample of survey respondents, as well as across three subsamples based on household income ranges: less than $20,000, between $40,000 to $50,000, and greater than $70,000. Reported unpleasantness ranges in value from 1 to 7, with higher values corresponding to higher levels of unpleasantness.

TABLE 9 Unpleasantness of budgeting by income predictability.

	(1)	(2)
Quartile 1	0.81***	0.88***
	(0.07)	(0.08)
Quartile 2	0.41***	0.55***
	(0.08)	(0.08)
Quartile 3	0.18*	0.25***
	(0.08)	(0.08)
Constant	3.38***	3.31***
	(0.05)	(0.05)
Demographic controls		Y
Household structure controls		Y
Income controls		Y
N	3781	3781

Note: This table presents estimates of the differential in reported unpleasant- ness associated with budgeting for respondents in the first, second, or third quartile of income predictability relative to those in the highest quartile of income predictability (i.e., those with the most predictable incomes). Each column reports estimates from an OLS regression where the dependent variable is the reported level of unpleasantness. Coefficients are reported for a "Quartile 1" indicator, "Quartile 2" indicator, and "Quartile 3" indicator, denoting whether a respondent is among those in the first, second, or third quartile of income predictability, respectively. The estimate for "Constant" in the bottom row gives the average baseline level of unpleasantness for the omitted group of respondents in the highest quartile of income predictability (Quartile 4). Column (1) presents un- adjusted estimates, and Column (2) presents regression-adjusted estimates that control for age, gender, education, employment status, marital status, number of children in the household, household income, and pay frequency. Significance levels 10%, 5%, 1%, and 0.1% are denoted by +, *, **, and ***, respectively.

Figure 9 presents the frequency of each of these reasons for respondents in the lowest quartile of income predictability. We find that for those with unpredictable income, any unpleasantness associated with budgeting appears to arise predominantly because of either how the budgeting process makes them feel (52.4%) or practical challenges to budgeting, such as the amount of time it takes (32.4%) or the difficulty in following a budget long-term (37.5%). Once again, we find evidence that suggests few respondents find budgeting not to be useful. Only a small share (11.7%) indicate that budgeting is unpleasant because it does not help them reach their financial goals. Taken together, the patterns we identify in this section indicate that for those with highly unpredictable income, solutions that help to address the practical challenges that make budgeting unpleasant or difficult may be more promising than policies aimed at increasing awareness of the benefits of budgeting.

4 | CONCLUSION

Unpredictable income fluctuations pose a significant threat to economic stability for many U.S. households. While budgeting may help to mitigate some of the negative effects associated with income unpredictability, the process of budgeting can also be more challenging when income is unpredictable. This paper uses cross-sectional data from a nationally-representative panel of nearly 4000 U.S. adults to provide new evidence on the relationship between income predictability, financial wellbeing, and budgeting behavior. We find evidence of a strong

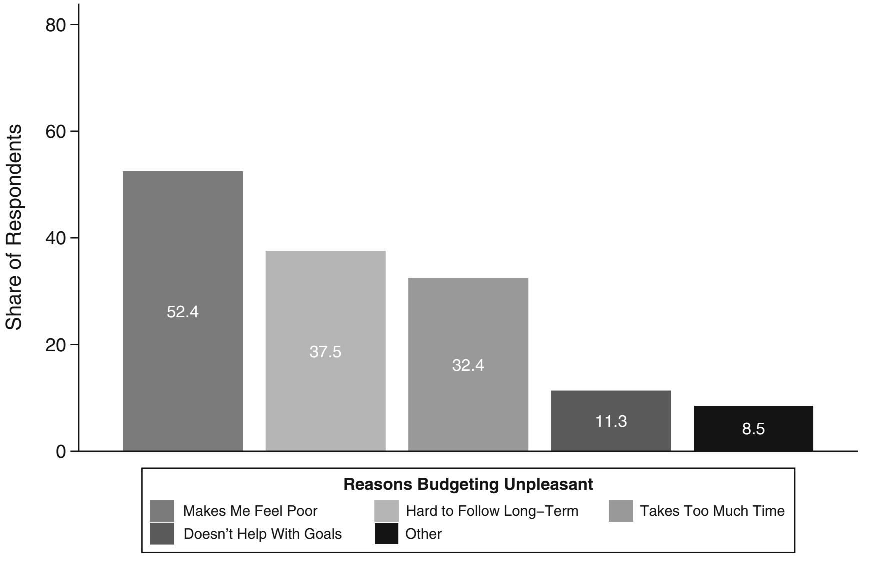

FIGURE 9 Reasons budgeting unpleasant for those with most unpredictable income. This figure plots the distribution of the share of respondents in the lowest quartile of income predictability (i.e., those with the most unpredictable incomes) that find budgeting unpleasant who indicate a particular reason why budgeting is unpleasant for each of the following set of potential motivations: takes too much time, too hard to follow long-term, makes me feel like I have less money than I thought I did, and doesn't help me reach my financial goals. In addition to these response options, respondents could provide their own reasoning through an open-ended response ("other").

negative relationship between income unpredictability and financial wellbeing. Households with unpredictable incomes have substantially lower levels of financial wellbeing, even after conditioning on income, pay frequency, and a host of demographic characteristics. The strength of this negative relationship is, however, substantially attenuated for those who budget. This suggests that budgeting may indeed be a useful tool for mitigating the adverse effects of unpredictable income. Nonetheless, despite these benefits, we find a significantly lower propensity to budget among those with the most unpredictable income.

These results underscore a fundamental challenge for practitioners and policymakers: the households who might benefit most from budgeting are also the ones least likely to engage in budgeting. Our findings suggest that practical challenges, such as the time commitment required for budgeting or the difficulty in following a budget over the long-term, rather than a lack of interest or motivation may be the primary barriers to budgeting for these households. These results also add an additional layer to prior research that has demonstrated how limited resources can place an increased cognitive burden on individual's mental resources, making it more difficult to engage in complex decision-making and put effort towards tasks such as financial planning (Mullainathan & Shafir, 2013; Shah et al., 2012). Our findings suggest that it is not only the lack of resources but also the unpredictability in the availability of those limited resources that can impede efforts to manage one's finances.

Taken together, our research suggests several potential implications for helping households to weather unpredictable income. As a practical implication, our findings underscore the need for financial education programs and counselors to directly address income unpredictability as a key challenge to household financial management strategies, taking care to choose approaches that avoid placing blame on the individual. One of the difficulties practitioners face in navigating these conversations is that financial uncertainty can undermine the ability for individuals to build their sense of confidence and self-efficacy in managing their finances (Peetz et al., 2021; Plunkett & Buehner, 2007). Given the particular barriers to budgeting that we observe, financial counselors should consider more targeted approaches that might help make budgeting less time-consuming or easier to follow long-term for those with unpredictable incomes. For example, recent work has shown that strategies that help make atypical expenses more salient may help improve the accuracy of predictions of future expenses, which in turn may improve the efficacy of budgeting efforts (Howard et al., 2022).

In light of the challenges to encouraging those with unpredictable incomes to budget, our findings indicate the need for policies that also address the underlying structural issues that contribute to income volatility and unpredictability or ways to help insure households against such circumstances. A growing body of evidence has documented increasing precarity in the labor market, particularly with the rise of part-time work, on-demand scheduling, and gig work (Garin et al., 2022; Harknett et al., 2021; Lambert et al., 2019; Sayre, 2023; Schneider & Harknett, 2019; Schneider & Harknett, 2021), and how this precarity can contribute to the income uncertainty households face. Our findings underscore that potential benefit of regulatory efforts, such as the recent adoption of fair work scheduling laws in several states and municipalities, that can help improve the predictability of household incomes.

One of the key contributions of our paper is the use of a new survey dataset that allows us to directly link income predictability with budgeting behavior. However, an important caveat to our analysis is that the relationships we examine are necessarily descriptive in nature. While we attempt to control for key demographic and economic characteristics, it may be that there are important unobservable traits or environmental features that correlate with both income predictability and our outcomes of interest. For example, prior research has shown that households with a low propensity to plan are less likely to budget (Ameriks et al., 2003). It may be that those with a low propensity to plan are also less likely to seek out jobs that pay predictable income. It is our hope that future work is able to study exogenous shifts in income predictability in a way that allows for more direct causal inference on the relationship between income predictability and budgeting.

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ENDNOTES

[1] In general, there is very limited evidence on the causal relationship between budgeting and financial outcomes. Lukas and Howard (2023) provide some evidence from a field experiment showing that setting a budget leads to lower discretionary spending. However, the causal effects of budgeting on broader measures of financial outcomes and wellbeing remains an important area for future research.

[2] For example, recent work has highlighted how scheduling unpredictability is associated with increased usage of high-cost debt (Amorim & Schneider, 2022), greater material hardship (Schneider & Harknett, 2021), and lower overall wellbeing (Peetz & Robson, 2019).

[3] To distinguish between formal and informal budgeting, the survey asks respondents the following question: "Do you keep your budget formally (for example, written down or on a website) or informally (for example, keep it in your head)?"

[4] In Table C1, we compare our survey measure of income predictability to the measures of in- come volatility from the National Financial Well-Being Survey and SHED and show how each measure of predictability or

volatility varies with key demographic and economics characteristics. Column (1) of Table C1 uses our measure of income predictability that ranges from 1 to 7, with higher scores denoting more predictable incomes, and repeats the summary statistics presented in column (1) of Table 2. For the National Financial Well-Being Survey and SHED (columns (2) and (3)), income volatility is measured on a scale ranging from 1 to 3, with 3 being the most volatile (in reverse order of our income predictability measure). We find consistent patterns across all three measures.

[5] In Table B1, we also report results from an OLS regression that uses midpoint values of the asset level ranges as the outcome. We find similar results to that of Table 3.

[6] In Table B2, we also report results from an OLS regression that uses midpoint values of the savings level ranges as the outcome. We find similar results to that of Table 4.

[7] The regressions in Table 5 assume linearity and provide valid estimates of the conditional expectation function. However, given the categorical nature of our measure of confidence, we also run our analysis using an ordered logit. As Table B3 shows, we find similar results.

[8] In the Appendix, we provide additional details on the construction of the composite index of perceived wellbeing. We also examine the relationship between income predictability and each of the four individual questions separately and find similar results to that of our main analysis.

[9] Interestingly, the nature of the budgets that households keep also appears to vary systematically with income predictability. In particular, in Appendix E we document that, conditional on budgeting, those with unpredictable incomes appear to maintain budgets that rely less heavily on explicit categorizations of spending (e.g., groceries versus gas) and, to the extent that they do track spending within such categories, their categorizations are more malleable.

[10] The original survey question asks respondents to report how unpleasant they find keeping a budget on a seven-point scale that ranges from "1: I find keeping a budget very unpleasant" to "7: I find keeping a budget very fun." For ease of exposition, we reverse the scale so that higher scores are associated with higher levels of unpleasantness. For those currently not budgeting, we asked respondents to answer either thinking of their previous budget or, if they had never budgeted before, the budget they would keep if they were to start budgeting.

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APPENDIX A: SAMPLE REPRESENTATIVENESS

TABLE A1 Benchmarking exercise (Zhang et al., 2022).

	(1) Survey data	(2) ACS 2010	(3) ACS 2018	(4) CES
Age bracket				
18–24 years	10.2	13.1	12.4	—
25–34 years	20.9	17.5	17.9	—
35–44 years	19.8	18.4	16.3	—
45–54 years	17.7	19.3	17.1	—
55+ years	31.4	31.8	36.3	—
Female	53.5	51.5	51.3	—
Employment status				
Employed	56.3	59.4	59.3	—
Unemployed or not in labor force	43.6	40.6	42.6	—
Education				
High-school degree or less	24.8	44.5	40.0	—
Some college or Associate's degree	38.3	30.0	31.0	—
Bachelor's degree or more	36.9	25.4	29.0	—
Household income brackets				
<20 K	17.7	23.6	15.5	—
20 K–50 K	33.9	24.7	26.6	—
50 K–70/75 K	17.8	18.6	17.5	—
>70/75 K	30.7	33.2	40.4	—
Pay frequency				
Weekly	25.4	—	—	33.8
Semi-monthly	18.5	—	—	18.6
Biweekly	36.2	—	—	42.2
Monthly	14.2	—	—	5.4
Irregular pay or other	5.7	—	—	—

Note: This table replicates Table A1 from Zhang et al. (2022) and compares the distribution of individuals in the survey data (column (1)) along key characteristics to the 2010 American Community Survey (ACS) 5-Year Estimates (column (2)), the 2018 American Community Survey (ACS) 5-Year Estimates (column (3)) and the February 2019 U.S. Bureau of Labor Statistics Current Employment Statistics (CES) survey (column (4)). ACS estimates for employment status are based on the population of U.S. residents 16 and older, not 18 and older as in the survey data. The top two household income brackets for the survey data represent those with incomes between $50,000 to $70,000 and $70,000 or more. The top two household income brackets for the ACS data represent those with incomes between $50,000 to $75,000 and $75,000 or more.

APPENDIX B: ROBUSTNESS CHECK

TABLE B1 Asset levels by income predictability–midpoint values.

	(1)	(2)
Quartile 1	−22,891.81***	−9069.17***
	(1670.72)	(1483.05)
Quartile 2	−12,643.89***	−6504.02***
	(1736.37)	(1473.76)
Quartile 3	−1781.83	−1919.00
	(1688.66)	(1391.75)
Constant	36,931.44***	32,192.80***
	(1130.23)	(956.25)
Demographic controls		Y
Household structure controls		Y
Income controls		Y
N	3777	3777

Note: This table presents estimates of the differential in assets for respondents in the first, second, or third quartile of income predictability relative to those in the highest quartile of income predictability (i.e., those with the most predictable incomes). Households were able to indicate which of the following ranges aligns with the amount of assets they hold: $0–$249, $250–$499, $500–$999, $1000–$4999, $5000–$9999, $10,000–$99,999, and $100,000+. Each column reports estimates from an OLS regression that uses the midpoint of the re- ported asset level range as the dependent variable. For the highest asset range, which is unbounded from above, the dependent variable is set equal to the lower bound of the range ($100,000). Coefficients are reported for a "Quartile 1" indicator, "Quartile 2" indicator, and "Quartile 3" indicator, denoting whether a respondent is among those in the first, second, or third quartile of income predictability, respectively. The estimate for "Constant" in the bottom row gives the average baseline amount of assets for the omitted group of respondents in the highest quartile of income predictability (Quartile 4). Column (1) presents unadjusted estimates, and Column (2) presents regression-adjusted estimates that control for age, gender, education, employment status, marital status, number of children in the household, household income, and pay frequency. Significance levels 10%, 5%, 1%, and 0.1% are denoted by +, *, **, and ***, respectively.

TABLE B2 How much saved last month by income predictability—Midpoint values.

	(1)	(2)
Quartile 1	−154.01***	−83.54***
	(15.17)	(14.78)
Quartile 2	−75.78***	−59.07***
	(15.52)	(14.42)
Quartile 3	−29.78*	−34.19*
	(15.02)	(13.52)
Constant	362.54***	342.94***
	(10.26)	(9.50)
Demographic controls		Y
Household structure controls		Y
Income controls		Y
N	3049	3049

Note: This table presents estimates of the differential in the amount saved in the prior month for respondents in the first, second, or third quartile of income predictability relative to those in the highest quartile of income predictability (i.e., those with the most predictable incomes). Households were able to indicate which of the following ranges aligns with the amount they saved in the prior month: $0, $1–$49, $50–$99, $100–$499, $500–$999, and $1000+. Each column reports estimates from an OLS regression that uses the midpoint of the reported savings level range as the dependent variable. For the highest savings range, which is unbounded from above, the dependent variable is set equal to the lower bound of the range ($1000). Coefficients are reported for a "Quartile 1" indicator, "Quartile 2" indicator, and "Quartile 3" indicator, denoting whether a respondent is among those in the first, second, or third quartile of income predictability, respectively. The estimate for "Constant" in the bottom row gives the average baseline amount saved in the prior month for the omitted group of respondents in the highest quartile of income predictability (Quartile 4). Column (1) presents unadjusted estimates, and Column (2) presents regression-adjusted estimates that control for age, gender, education, employment status, marital status, number of children in the household, household income, and pay frequency. Significance levels 10%, 5%, 1%, and 0.1% are denoted by +, *, **, and ***, respectively.

TABLE B3 Confidence in ability to come up with funds by income predictability—Ordered logit.

	(1)	(2)	(3)	(4)
	$100	**$500**	**$1000**	**$5000**
Quartile 1	0.54***	0.55***	0.56***	0.54***
	(0.05)	(0.05)	(0.05)	(0.05)
Quartile 2	0.77**	0.71***	0.72***	0.68***
	(0.08)	(0.06)	(0.07)	(0.06)
Quartile 3	1.05	0.99	0.90	0.84+
	(0.10)	(0.09)	(0.08)	(0.07)
Demographic controls	Y	Y	Y	Y
Household structure controls	Y	Y	Y	Y
Income controls	Y	Y	Y	Y
N	3781	3781	3781	3781

Note: This table presents odds ratio estimates of the differential confidence in the ability to come up with funds for respondents in the first, second, or third quartile of income predictability relative to those in the highest quartile of income predictability (i.e., those with the most predictable incomes). Each column reports estimates from an ordered logit regression where the dependent variable is the reported confidence in the ability to come up with the amount of funds indicated in the column heading if an unexpected need arose. Controls include age, gender, education, employment status, marital status, number of children in the household, household income, and pay frequency. Coefficients are reported for a "Quartile 1" indicator, "Quartile 2" indicator, and "Quartile 3" indicator, denoting whether a respondent is among those in the first, second, or third quartile of income predictability, respectively. The omitted group are respondents in the highest quartile of income predictability (Quartile 4). Significance levels 10%, 5%, 1%, and 0.1% are denoted by +, *, **, and ***, respectively.

APPENDIX C: INCOME PREDICTABILITY VERSUS VOLATILITY

TABLE C1 Income predictability versus volatility by individual characteristics.

	Predictability	Volatility	
	(1)	(2)	(3)
	Survey	CFPB	SHED
	Mean score	Mean score	Mean score
Overall sample	5.30	1.34	1.35
Panel A. Demographic characteristics			
Age bracket			
18–24 years	4.54	1.42	1.67
25–34 years	5.01	1.42	1.45
35–44 years	5.16	1.41	1.39
45–54 years	5.31	1.38	1.35
55+ years	5.83	1.26	1.28
Gender			
Female	5.23	1.38	1.35
Male	5.39	1.30	1.35
Employment status[a]			
Student	4.59	1.36	1.60
Part-time	4.72	1.52	1.53
Full-time	5.35	1.30	1.28
Not currently employed	5.55	1.28	1.24
Education			
High-school degree or less	5.05	1.43	1.38
Some college or Associate's degree	5.22	1.37	1.37
Bachelor's degree or more	5.56	1.28	1.31
Marital status			
Lives with spouse/partner	5.20	1.36	1.38
Does not live with spouse/partner	5.39	1.33	1.33
Children in the household			
Does not have children	5.35	1.31	1.34
Has children	5.23	1.39	1.37
Panel B. Financial characteristics			
Household income			
<20 K	4.90	1.51	1.44
20 K–30 K	5.10	1.48	1.40
30 K–40 K	5.22	1.37	1.34

(Continues)

TABLE C1 (Continued)

	Predictability	Volatility	
	(1)	(2)	(3)
	Survey	CFPB	SHED
	Mean score	Mean score	Mean score
40K–50 K	5.22	1.34	1.39
50 K–60 K	5.39	1.29	1.33
60 K–70 K[b]	5.37	1.32	1.33
>70 K[b]	5.63	1.28	1.31
Pay frequency			
Weekly	5.04	—	—
Biweekly	5.29	—	—
Semi-monthly	5.44	—	—
Monthly	5.86	—	—
Irregular frequency	3.62	—	—

Note: This table compares our survey measure of income predictability (column (1)) to measures of income volatility from the Consumer Financial Protection Bureau (CFPB) National Financial Well-Being Survey (column (2)) and the 2018 Survey of Household Economics and Decision-making (column (3)). Each column shows how the indicated measure of predictability or volatility varies with key demographic and economics characteristics. For the CFPB and SHED surveys, income volatility is measured on a scale ranging from 1 to 3, with 3 being the most volatile. For our measure of predictability, the scale ranges from 1 to 7 and the ordering is reversed, so that 7 denotes the most predictable incomes.

[a]The mean income volatility scores of part-time and full-time employed respondents in the CFPB data (column (2)) and the SHED (column (3)) do not include those that are self-employed.

[b]The top two household income brackets for our survey data represent those with incomes between $50,000 to $70,000 and $70,000 or more. The top two household income brackets for the CFPB data (column (2)) and the SHED (column (3)) represent those with incomes between $50,000 and $75,000 and $75,000 or more.

APPENDIX D: PERCEIVED WELLBEING INDEX

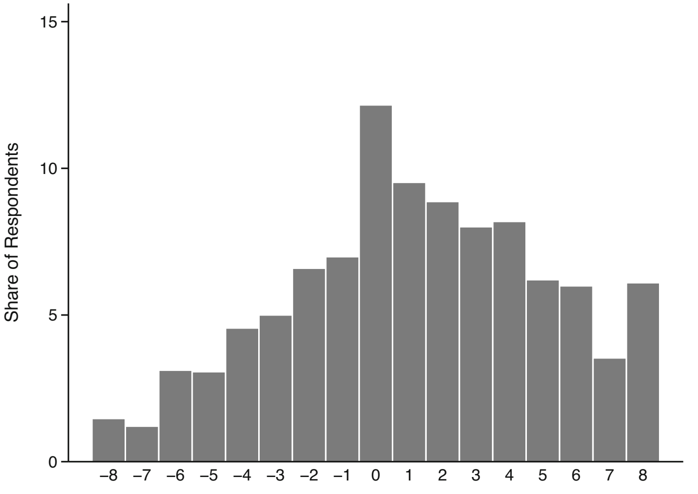

FIGURE D1 Distribution of perceived wellbeing index scores. This figure plots the distribution of the perceived wellbeing index scores for the full sample of survey respondents. The index is a composite score created by combining the individual scores from responses to four questions: (1) I will achieve the financial goals that I have set for myself; (2) I have saved (or will be able to save) enough money to last me to the end of my life; (3) I am behind on my finances; and (4) Because of my money situation, I feel I will never have the things I want in life. Responses for each question are measured using a five-point scale that ranges from "1: Describes me not at all" to "5: Describes me completely." The first two questions are designed to capture respondents' expectations about their future financial security whereas the latter two questions are designed to capture respondents' stress about their current money management (Netemeyer et al., 2018). To create the index, we add the scores for the two questions on expected future financial security with the negative of scores for the latter two questions on current money management stress. The index ranges in value from −8 to 8, with higher values indicating higher perceived wellbeing.

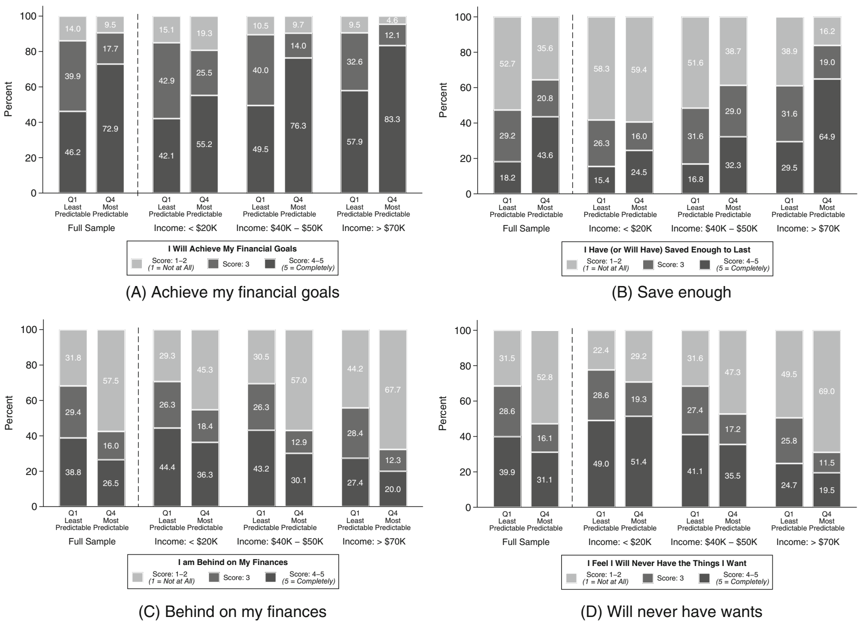

FIGURE D2 Individual perceived financial wellbeing questions by income predictability. This figure plots the distribution of perceived wellbeing scores for those in the lowest (Q1) versus highest (Q4) quartile of income predictability for the full sample of survey respondents, as well as across three subsamples based on household income ranges: less than \$20,000, between \$40,000 to \$50,000, and greater than \$70,000. Scores are shown for responses to four questions: (1) I will achieve the financial goals that I have set for myself (Panel A); (2) I have saved (or will be able to save) enough money to last me to the end of my life (Panel B); (3) I am behind on my finances (Panel C); and (4) Because of my money situation, I feel I will never have the things I want in life (Panel D). Responses for each question are measured using a five-point scale that ranges from "1: Describes me not at all" to "5: Describes me completely."

TABLE D1 Perceived financial wellbeing by income predictability.

	I will achieve my financial goals		I will be able to save enough $ for lifetime	
	(1)	(2)	(3)	(4)
Panel A. Expected future financial security				
Quartile 1	−0.58***	−0.61***	−0.69***	−0.56***
	(0.04)	(0.05)	(0.06)	(0.06)
Quartile 2	−0.28***	−0.36***	−0.30***	−0.31***
	(0.05)	(0.05)	(0.06)	(0.06)
Quartile 3	−0.13**	−0.18***	−0.08	−0.14*
	(0.04)	(0.04)	(0.06)	(0.06)
Constant	4.04***	4.07***	3.09***	3.07***
	(0.03)	(0.03)	(0.04)	(0.04)
Demographic controls		Y		Y
Household structure controls		Y		Y
Income controls		Y		Y
N	3781	3781	3781	3781
	I'm behind with my finances		**Because of $, i'll never have things i want**	
	(1)	**(2)**	**(3)**	**(4)**
Panel B. Current money management stress				
Quartile 1	0.67***	0.47***	0.56***	0.41***
	(0.06)	(0.06)	(0.06)	(0.06)
Quartile 2	0.28***	0.17**	0.21***	0.19**
	(0.06)	(0.06)	(0.06)	(0.06)
Quartile 3	0.09	0.06	0.04	0.08
	(0.06)	(0.06)	(0.06)	(0.06)
Constant	2.42***	2.51***	2.59***	2.63***
	(0.04)	(0.04)	(0.04)	(0.04)
Demographic controls		Y		Y
Household structure controls		Y		Y
Income controls		Y		Y
N	3781	3781	3781	3781

Note: This table presents estimates of the differential in perceived wellbeing for respondents in the first, second, or third quartile of income predictability relative to those in the highest quartile of income predictability (i.e., those with the most predictable incomes). Each column reports estimates from an OLS regression where the dependent variable is the respondent's perceived wellbeing as measured on a five-point scale for one of four questions: (1) I will achieve the financial goals that I have set for myself (columns (1) and (2) of Panel A); (2) I have saved (or will be able to save) enough money to last me to the end of my life (columns (3) and (4) of Panel A); (3) I am behind on my finances (columns (1) and (2) of Panel B); and (4) Because of my money situation, I feel I will never have the things I want in life (columns (3) and (4) of Panel B). Coefficients are reported for a “Quartile 1” indicator, “Quartile 2” indicator, and “Quartile 3” indicator, denoting whether a respondent is among those in the first, second, or third quartile of income predictability, respectively. The estimate for “Constant” in the bottom row gives the average baseline perceived wellbeing index score for the omitted group of respondents in the highest quartile of income predictability (Quartile 4). Columns (1) and (3) present unadjusted estimates, and Columns (2) and (4) present regression-adjusted estimates that control for age, gender, education, employment status, marital status, number of children in the household, household income, and pay frequency. Significance levels 10%, 5%, 1%, and 0.1% are denoted by +, *, **, and ***, respectively.

APPENDIX E: INCOME PREDICTABILITY AND BUDGET CATEGORIZATION

This section studies how the nature of the budgets that households keep varies with the predictability of their income. In particular, we explore whether there are differences in the extent to which individuals track their spending within distinct categories of consumption when budgeting and how the flexibility of these “budgetary categories” varies with income predictability. The use of budgetary categories is a common practice among those who budget (Zhang et al., 2022). Prior research has suggested that these categorizations are often malleable and that individuals may leverage this flexibility in order to rationalize spending (Cheema & Soman, 2006; Imas et al., 2021). In the case of individuals with unpredictable incomes, adhering to a more stringent use of budget categories may offer a useful framework for budgeting purposes. At the same time, the increased uncertainty stemming from unpredictable income may lead individuals to adopt a more flexible approach to budget categorization, allowing them to adapt to unexpected shocks to income.

To explore this further, we consider both how commonly individuals report using budgeting categories and the extent to which individuals change their budget categorizations from time to time (i.e., how malleable are these categories). Table E1 presents both unadjusted estimates (columns (1) and (3)) and regression-adjusted estimates controlling for demographics, household structure, income, and pay frequency (columns (2) and (4)) of the relationship between income predictability and these two outcomes of interest. The estimate for “Constant” in the bottom row gives the average likelihood a respondent in the highest quartile of income predictability reports using budgetary categories or treats these categories as malleable. The remaining estimates represent the relative difference in the propensity to use budgetary categories or to treat these categories as malleable for respondents in the indicated quartile of predictability.

We find that individuals with lower income predictability are significantly less likely to use budgetary categories relative to those in the highest quartile of income predictability. Furthermore, even when they do use budgetary categories, those with lower income predictability are significantly more likely to treat these categorizations as malleable. Taken together, these patterns suggest that individuals facing unpredictable income engage in more flexible budgeting practices relative to those with more predictable incomes.

TABLE E1 Budget categorization by income predictability.

	Prevalence of budget categories		Malleability of budget categories	
	(1)	**(2)**	**(3)**	**(4)**
Quartile 1	−0.06***	−0.04**	0.16***	0.15***
	(0.02)	(0.02)	(0.03)	(0.03)
Quartile 2	−0.01	−0.01	0.15***	0.13***
	(0.02)	(0.02)	(0.03)	(0.03)
Quartile 3	−0.01	−0.02	0.11***	0.09***
	(0.02)	(0.02)	(0.03)	(0.03)
Constant	0.93***	0.93***	0.36***	0.38***
	(0.01)	(0.01)	(0.02)	(0.02)
Demographic controls		Y		Y
Household structure controls		Y		Y
Income controls		Y		Y
N	2479	2479	2479	2479

Note: This table presents estimates of the differential in budget categorization for respondents in the first, second, or third quartile of income predictability relative to those in the highest quartile of income predictability (i.e., those with the most predictable incomes). The sample includes only those who report that they currently budget. Each column reports estimates from an OLS regression where the dependent variable is an indicator for whether the respondent uses budgetary categories (columns (1) and (2)) and an indicator for whether the respondent treats their budgetary categories as malleable (columns (3) and (4)). Coefficients are reported for a "Quartile 1" indicator, "Quartile 2" indicator, and "Quartile 3" indicator, denoting whether a respondent is among those in the first, second, or third quartile of income predictability, respectively. The estimate for "Constant" in the bottom row gives the average baseline likelihood of using budgetary categories or treating these categories as malleable for the omitted group of respondents in the highest quartile of income predictability (Quartile 4). Column (1) presents unadjusted estimates, and Column (2) presents regression-adjusted estimates that control for age, gender, education, employment status, marital status, number of children in the household, household income, and pay frequency. Significance levels 10%, 5%, 1%, and 0.1% are denoted by +, *, **, and ***, respectively.